

M51: Final Composite Dependent Variable

PCA on Raw Switch Counts (Time, Topic-LDA, Typing)

No standardization (z-score) applied

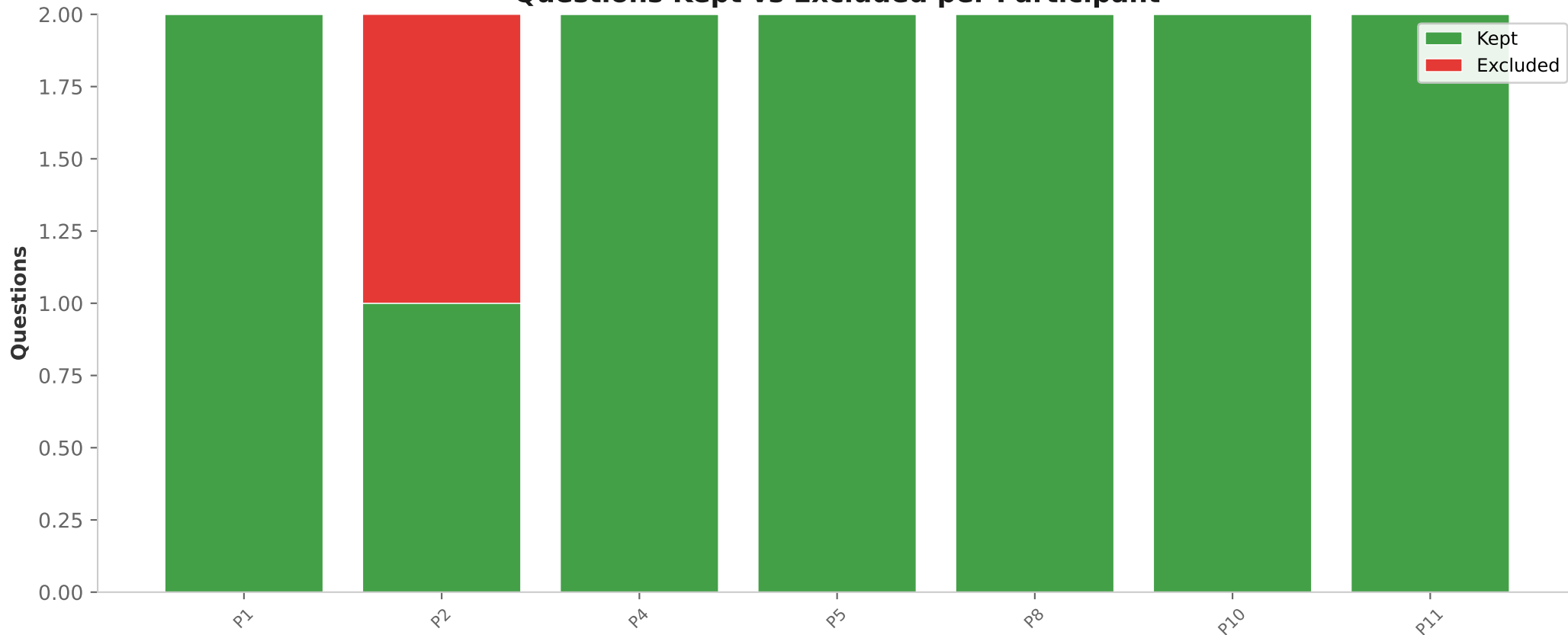
Exclusion Criteria Applied:

1. Questions with fewer than 3 page visits: 1 excluded
2. Questions with idle time $\geq 50\%$ after last meaningful event: 1 excluded
3. Participants > 3 SD from mean on any DV: 0 excluded

Result: 13 / 14 questions retained

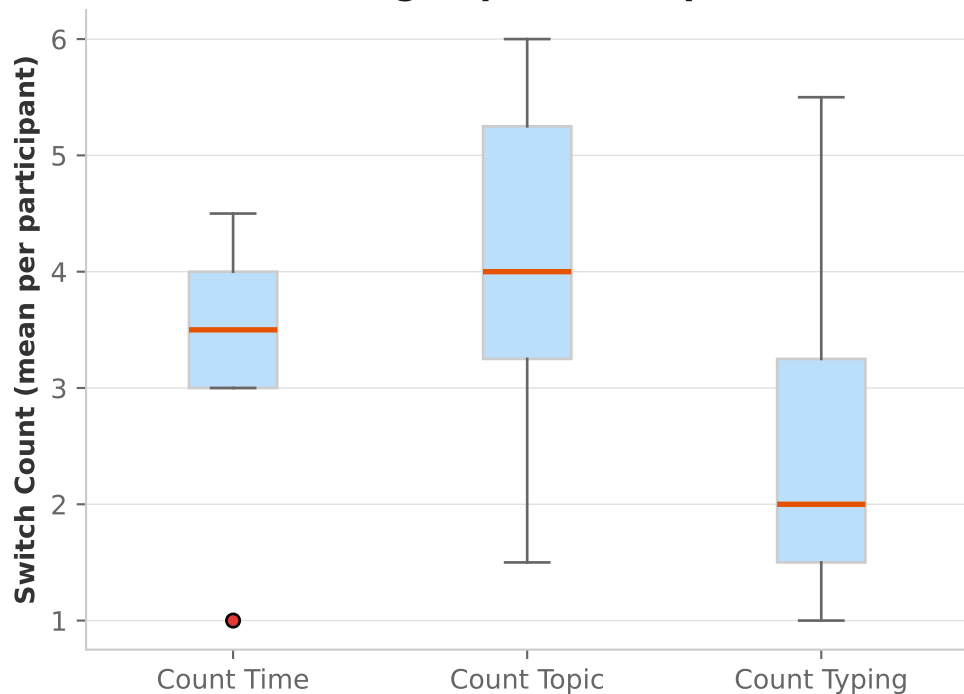
Final sample: N = 7 participants

Exclusion Summary: Questions Kept vs Excluded per Participant

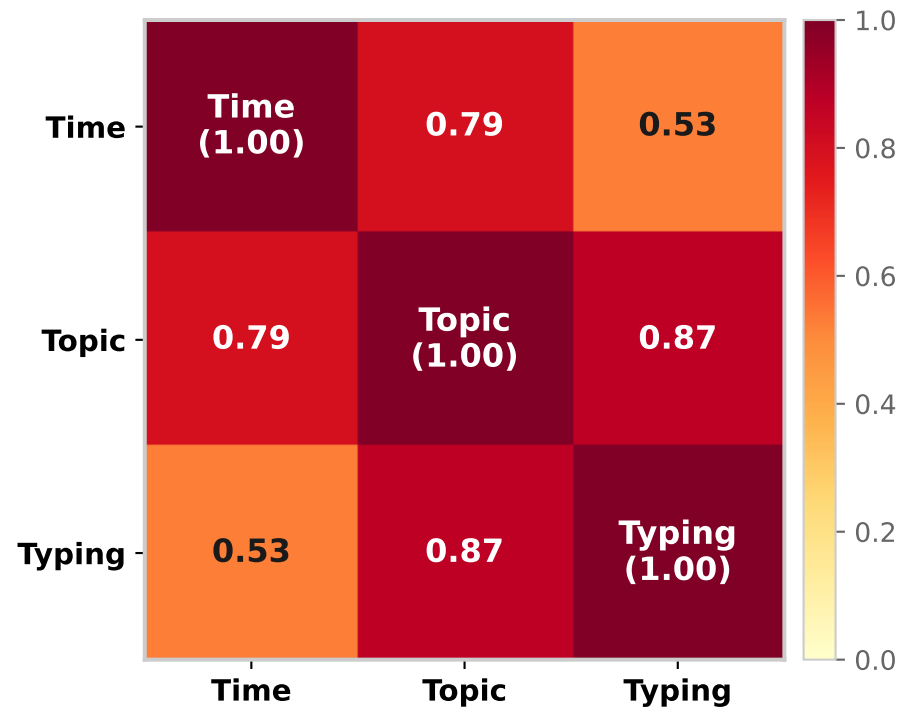


Descriptive Statistics: Raw Switch Counts

Distribution of Raw Switch Counts (Averaged per Participant)

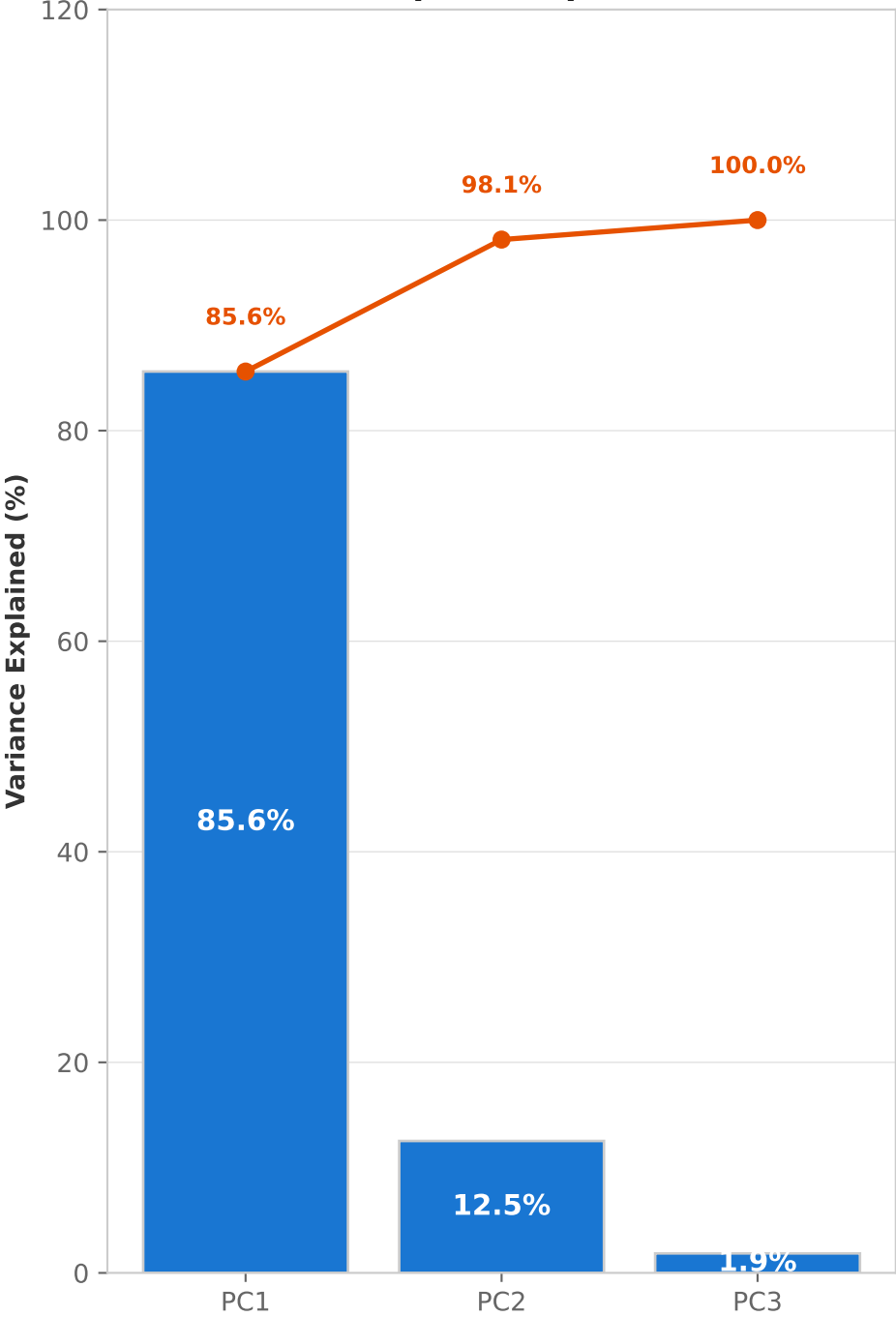


Correlation Between Switch Count Variables



PCA Results: Dimensionality Reduction on Raw Switch Counts

Variance Explained by Each Principal Component



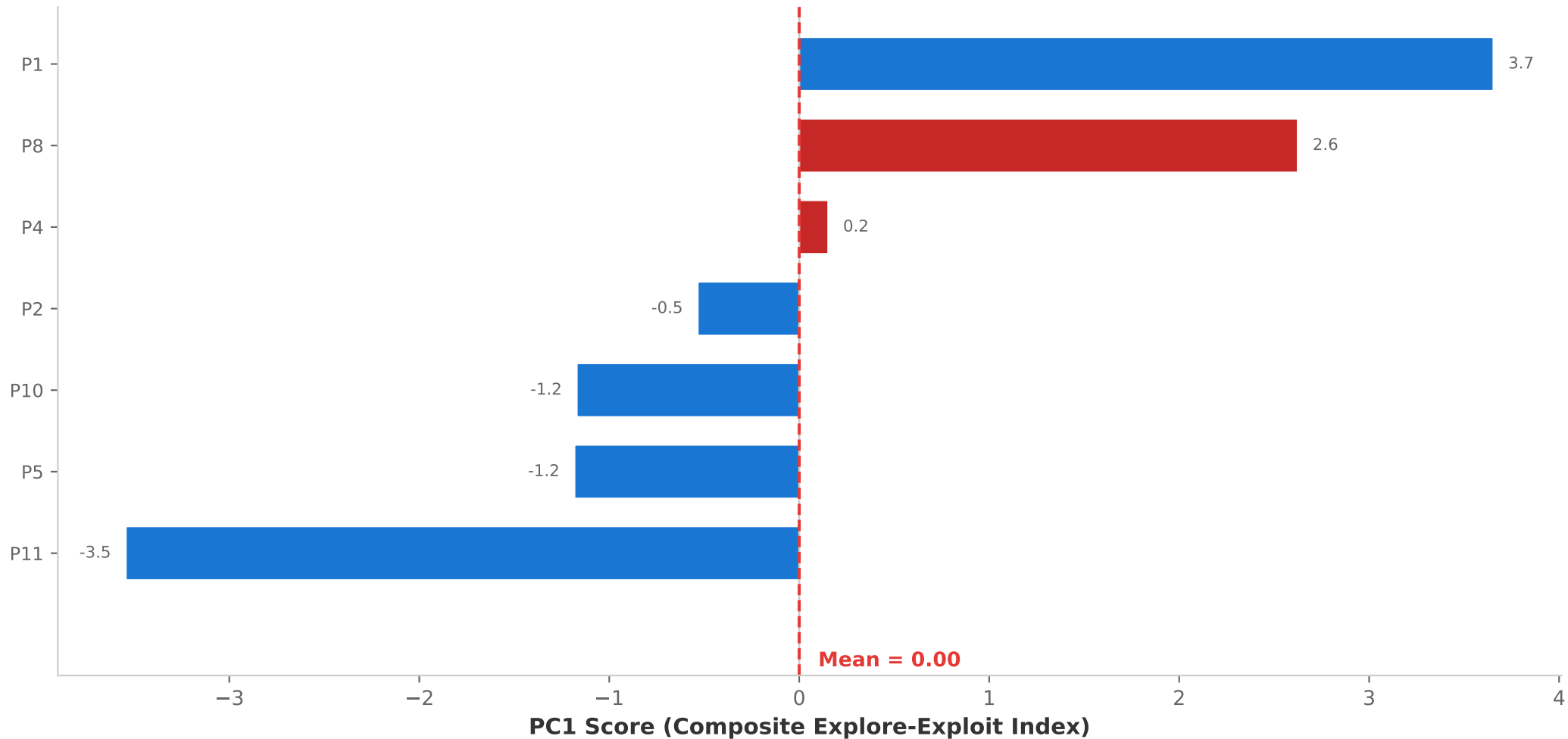
PCA Loadings (variable weights per component):

PC1 (85.6% variance):		
Count Typing	+0.671	+++++
Count Topic	+0.649	+++++
Count Time	+0.359	+++++
PC2 (12.5% variance):		
Count Time	+0.768	+++++
Count Typing	-0.607	-----
Count Topic	+0.203	++++
PC3 (1.9% variance):		
Count Topic	+0.733	+++++
Count Time	-0.530	-----
Count Typing	-0.426	-----

Interpretation:

- PC1: Primarily reflects typing switching (loading = +0.671)
Higher score = more typing transitions
- PC2: Primarily reflects time switching (loading = +0.768)
Higher score = more time transitions
- PC3: Primarily reflects topic switching (loading = +0.733)
Higher score = more topic transitions

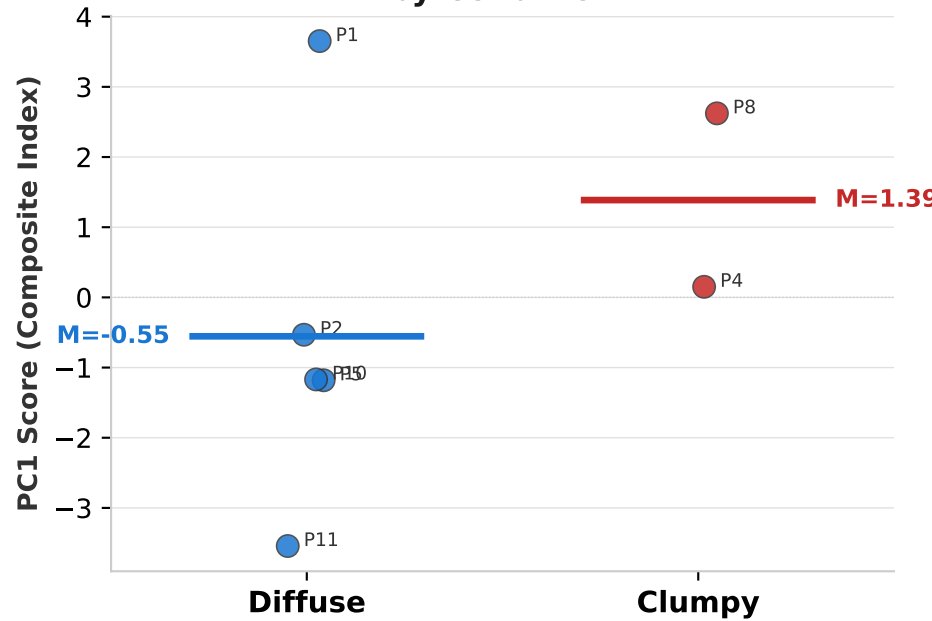
Individual PC1 Scores: Composite Explore-Exploit Index



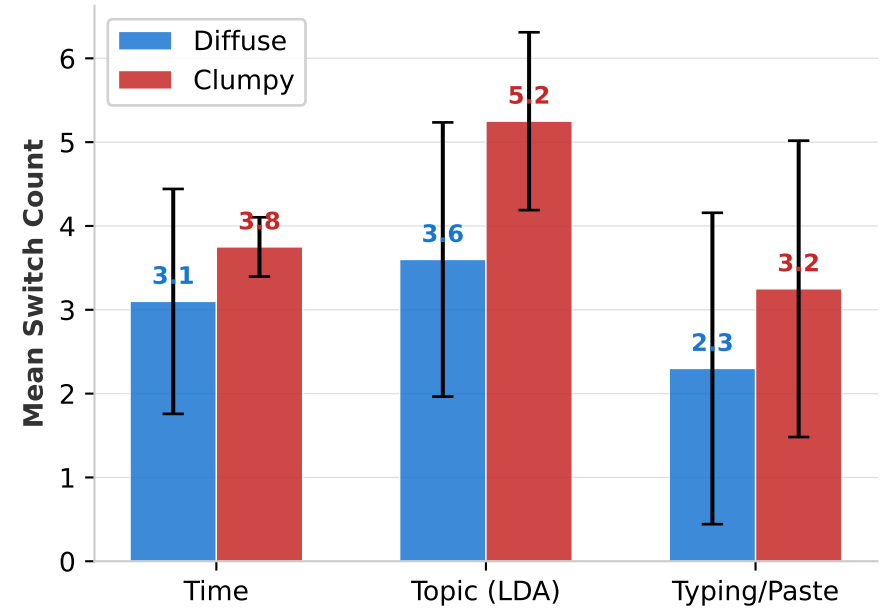
Condition Comparison: Clumpy vs Diffuse

N = 7 (Diffuse: 5, Clumpy: 2)

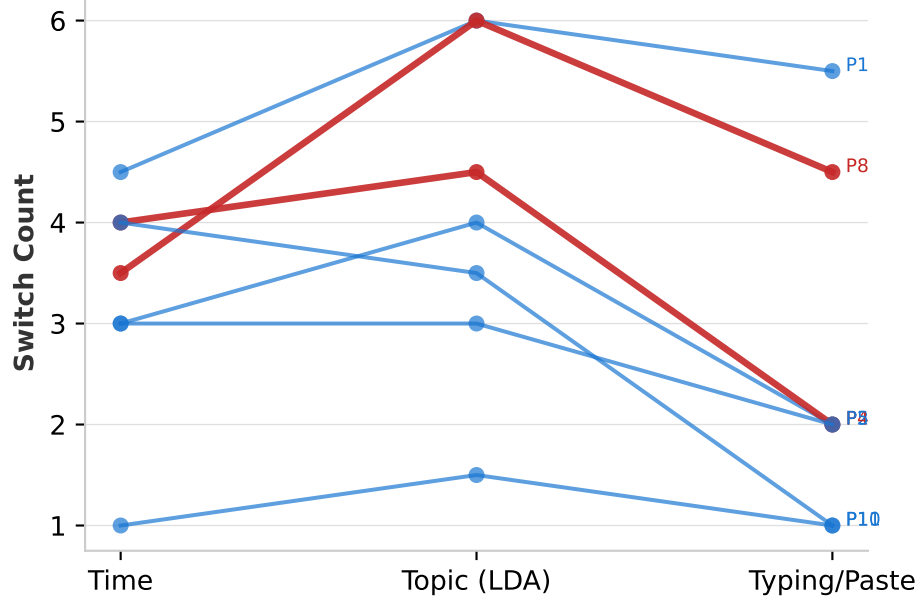
PC1: Composite Explore-Exploit
by Condition



Raw Switch Counts
by Condition (Mean +/- SD)



Individual Profiles
by Switch Type



Condition Comparison: Descriptive Statistics & t-Tests

N = 7 (Diffuse: 5, Clumpy: 2)

DESCRIPTIVE STATISTICS

Variable	Diffuse (n=5)	Clumpy (n=2)
Time	M = 3.10, SD = 1.34	M = 3.75, SD = 0.35
Topic (LDA)	M = 3.60, SD = 1.64	M = 5.25, SD = 1.06
Typing/Paste	M = 2.30, SD = 1.86	M = 3.25, SD = 1.77
PC1 (composite)	M = -0.55, SD = 2.62	M = 1.39, SD = 1.75

INDEPENDENT-SAMPLES t-TESTS (Welch)

Variable	t	df	p	Cohen d	Sig.
Time	-1.00	4.9	0.3640	+0.54	n.s.
Topic (LDA)	-1.58	3.1	0.2103	+1.07	n.s.
Typing/Paste	-0.63	2.0	0.5920	+0.52	n.s.
PC1 (composite)	-1.14	3.0	0.3370	+0.79	n.s.

* p < .05 ** p < .01 *** p < .001

INTERPRETATION

PC1 difference (Clumpy - Diffuse): +1.94
Clumpy condition shows more explore-exploit switches
Effect size (Cohen's d): +0.79
Result is NOT statistically significant (p = 0.3370)
Low statistical power due to small sample (N = 7)

Largest sub-variable effect: Topic (LDA)
d = +1.07, p = 0.2103

Note: Typing metric relies on paste events only
(answer_snapshot missing from current log).